



Statistical problems - data collection

Task A: Choosing what data to collect

- 1) A supermarket manager collects data to investigate ways to save money, increase sales and improve the customers' experience.

Match the data with the most appropriate tools for collecting it.

customers' opinions about a product that is for sale _____ questionnaire

number of customers visiting at each time of the day _____ stopwatch

weight of the vegetables which are left unpurchased _____ tally chart

time that customers spend queuing at the checkout _____ weighing scales

Task B: Choosing where to collect the data from

1) Jun wants to find out the average time people take to run a 100 metre race in his school. He assists in a PE lesson for a class in the youngest year group by timing how long each pupil takes to complete the race. He then calculates the mean time.

a) What is the **population** in Jun's investigation? *all the pupils in Jun's school*

b) Who does Jun use in his **sample**? *a class in the youngest year group*

c) Why could Jun's data be **biased**?

He has only collected data from a single age group.

Older pupils might run faster than younger pupils.

*The average time for the **sample** could be much higher than the average time for the whole school (assuming older students run faster).*

d) Propose an alternative way for Jun to collect his data.

*Use a **sample** that includes a mixture of people from each year group.*

The PE teachers may already have data for the whole school.

2) Laura wants to investigate how much time adults in the UK spend exercising, per week. She designs a questionnaire about people's exercise habits. She goes to her local gym and asks people to complete her questionnaire.

a) What is the **population** in Laura's investigation? *all adults in the UK*

b) Who does Laura use in her **sample**?

people who attend the gym that she went to who agree to take her questionnaire

c) Why could Laura's data be **biased**?

She has only collected data from people who are members of a gym.

People who go to a gym may do more exercise than people who don't.

Her results might suggest that adults in the UK spend more time doing exercise than they actually do.

d) Propose an alternative way for Laura to collect data using her questionnaire.

Conduct her questionnaire in a place where both gym members and non-gym members might go to (e.g. town high street or supermarket)

